# Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

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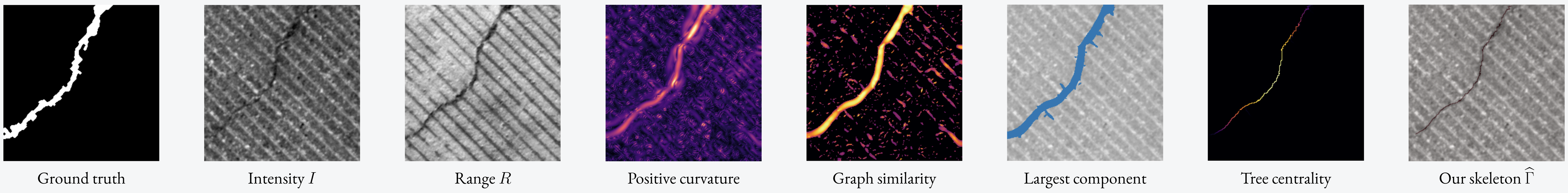
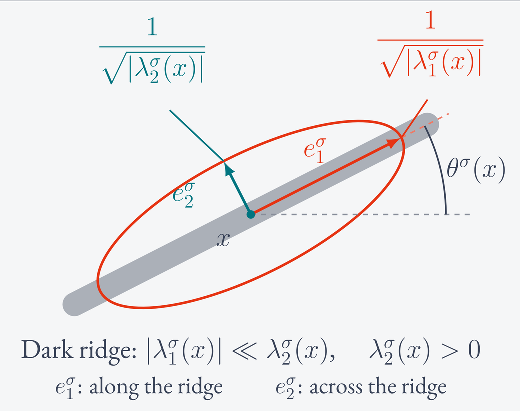
## From multimodal images to a crack skeleton




FIND example. Similarity: maximum over incident edges; skeleton in red.

## 1. Fuse the modalities

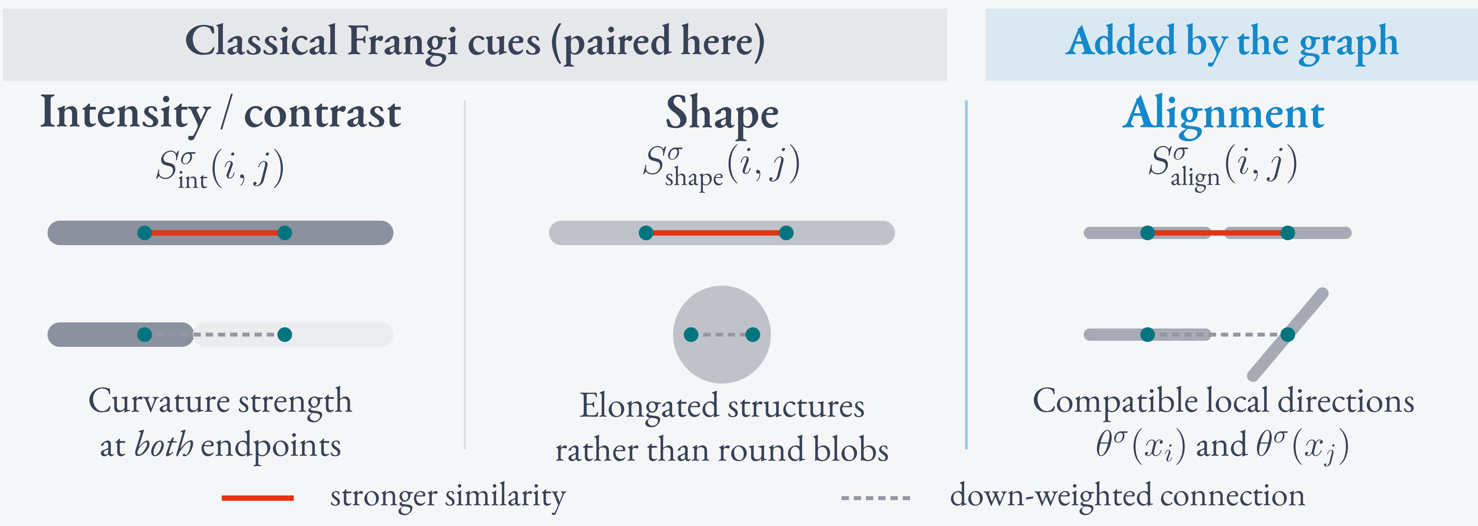



Hessian geometry [1] at scale $\sigma$: $|\lambda_1^\sigma| \leq |\lambda_2^\sigma|$; $\theta^\sigma$ follows the ridge.

$$\widehat{\mathcal{H}}_\sigma^{(m)}(x) = \frac{\mathcal{H}_\sigma^{(m)}(x)}{\max_{y\in\Omega} \left\| \mathcal{H}_\sigma^{(m)}(y) \right\|_2}, \qquad \mathcal{H}_\sigma^{\text{fused}} = \sum_{m\in\mathcal{M}} w_m \widehat{\mathcal{H}}_\sigma^{(m)}.$$

$w_m \geq 0, \sum_m w_m = 1$. Aligned dark-ridge inputs; eigensystem computed after fusion.

## 2. From pixel-wise Frangi [1] to pairwise links




**Classical Frangi cues + neighbor alignment.** For pixels with $0 < \rho_{ij} = \|x_i - x_j\|_2 \leq R$:

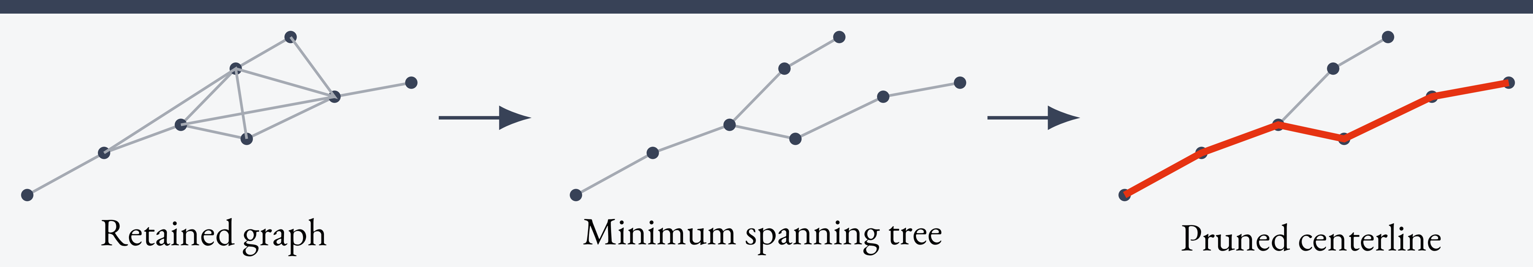
$$C_\sigma(x) = \max\{\lambda_2^\sigma(x), 0\}, \qquad R_B^\sigma(x) = \frac{|\lambda_1^\sigma(x)|}{|\lambda_2^\sigma(x)|}.$$

$$\boxed{S_{ij}^{(0)} = \max_{\sigma\in\Sigma} S_{\text{int}}^\sigma(i,j)\, S_{\text{shape}}^\sigma(i,j)\, S_{\text{align}}^\sigma(i,j)}$$

$$d_{ij} = \min\{1, \rho_{ij}(1 - S_{ij}^{(0)})\}, \qquad S_{ij} = 1 - d_{ij}.$$

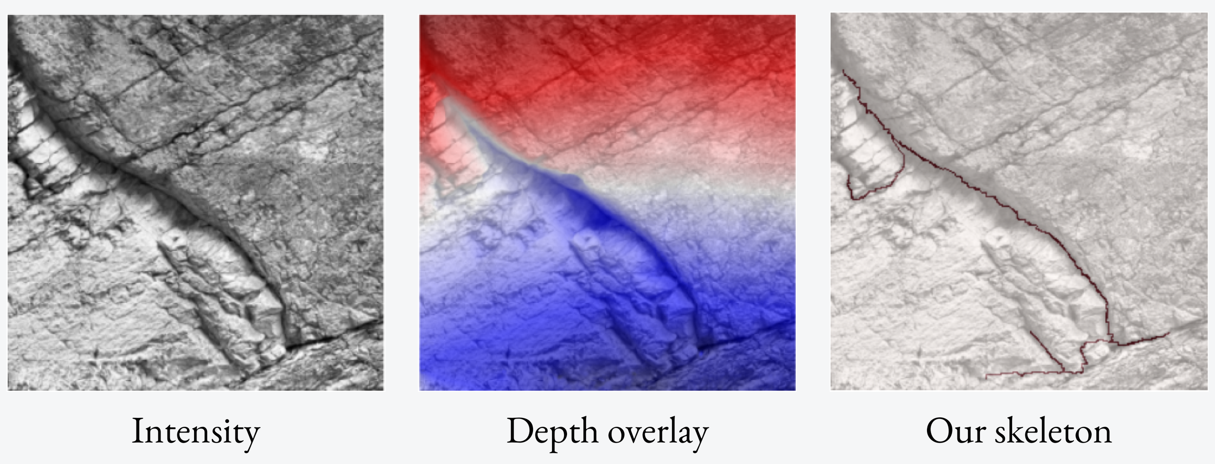
$\rho_{ij}$ penalizes long links; $\theta$ and $\theta + \pi$ denote the same axis.

## 3. Extract the centerline




Per component: minimum spanning tree with costs $d_{ij}$, then similarity-weighted centrality. FIND settings: $\Sigma = \{1, 3, 5, 7\}$, $R = 3$, retained fractions $\tau_E = \tau_V = 0.25$.

## Multimodality: Palais des Papes (France)




Fuse visible intensity and photogrammetric depth.

## 4. FIND: accuracy and noise

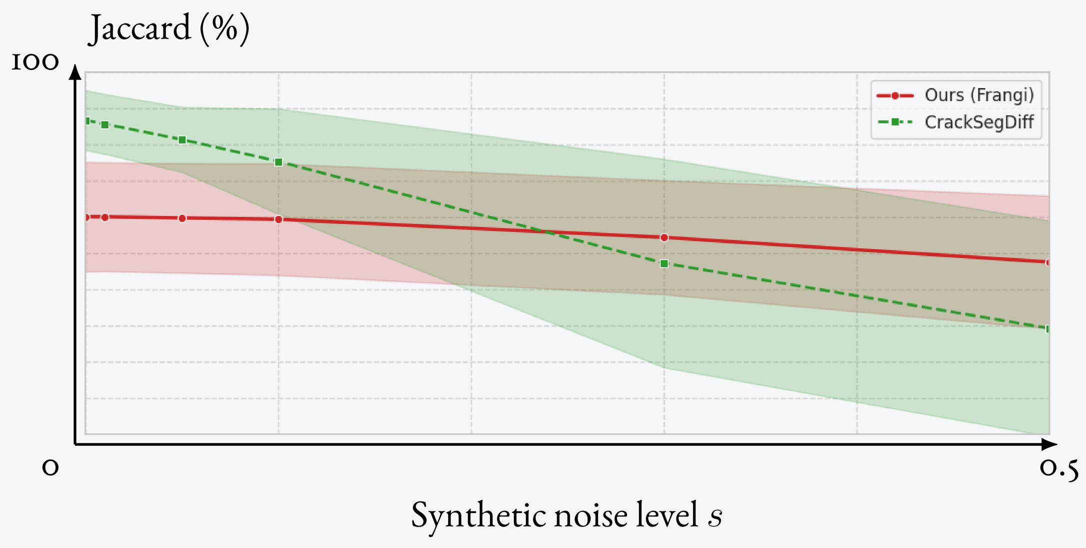
**Clean FIND: the supervised model performs best.**

| Method ($I + R$) | Jaccard ↑ | Tversky ↑ | $W$ ↓ |
|---|---|---|---|
| Ours, no training | 60% | 68% | 18 px |
| CrackSegDiff [3] | **87%** | **90%** | 5 px |

FIND [2], $256 \times 256$. Overlap: 3 px dilation of both skeletons; Tversky: $\alpha = 1, \beta = 0.5$. $W$: skeleton Wasserstein distance.

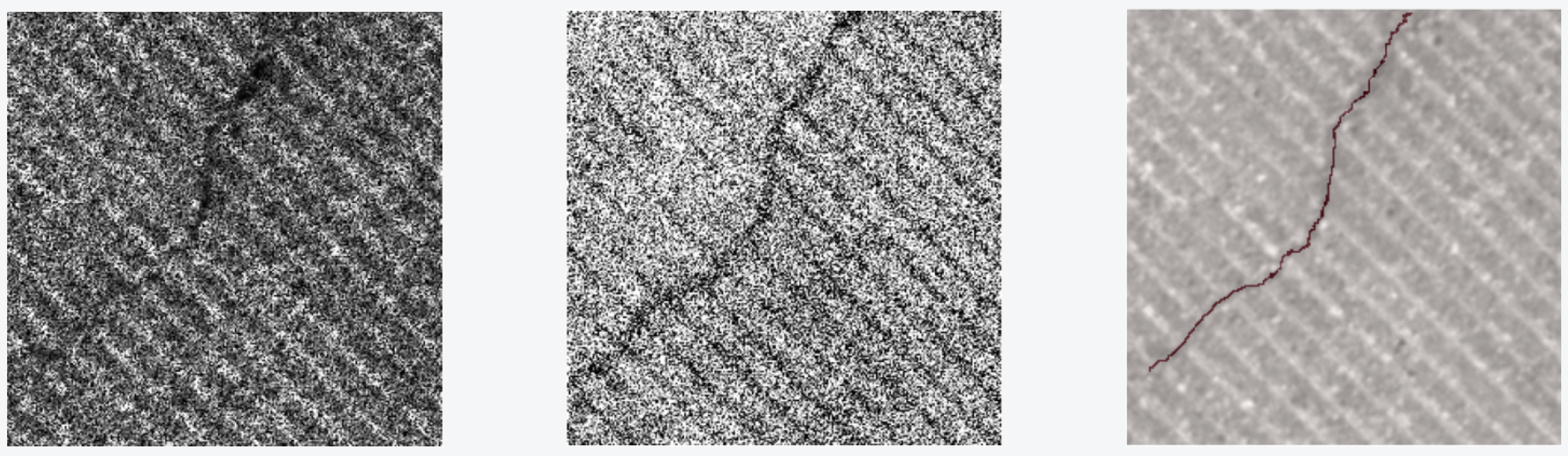
**Fusion improves Jaccard:** $I$: 41%; $I + R$: 60%; $I + R + F$: 63%. Equal weights; $F$ is filtered range.

**Slower degradation under the tested synthetic noise.**




Ours CrackSegDiff

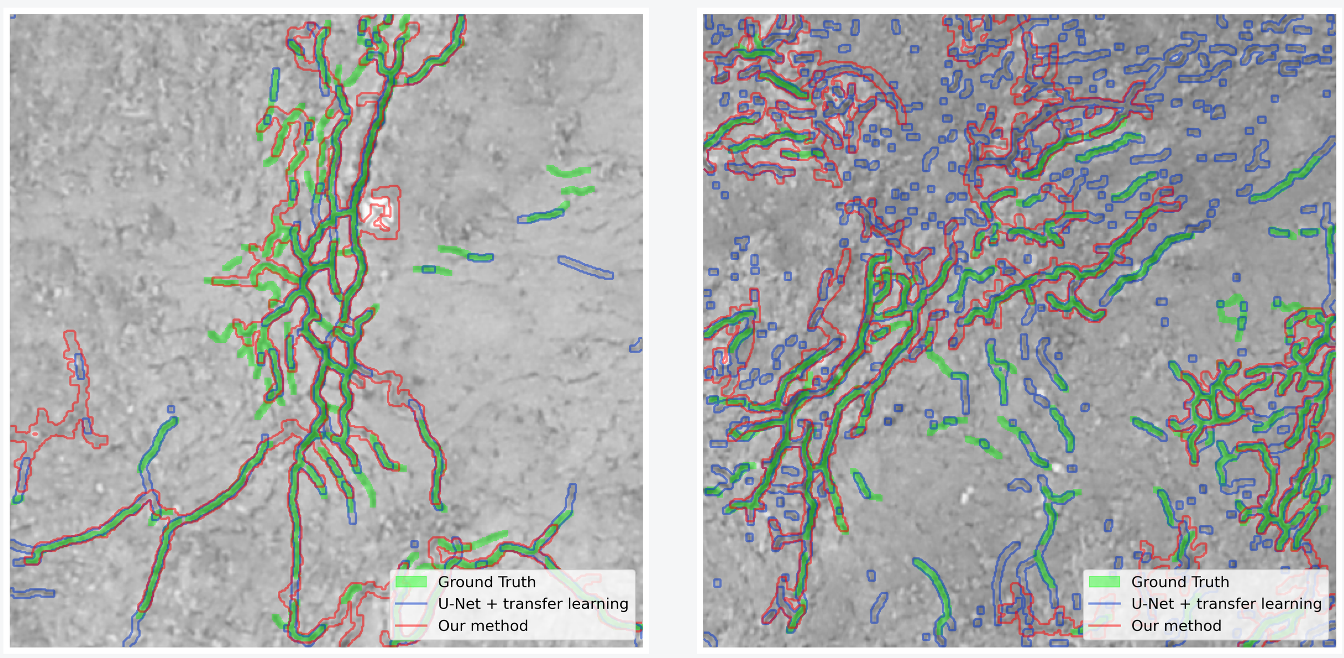
Bands: standard deviation.




$s = 0.5$: multiplicative speckle on $I$, additive Gaussian noise on $R$.

## 5. Geological domain shift [5]

**Vaches Noires.** Two Cerema UAV patches ($512 \times 512$).



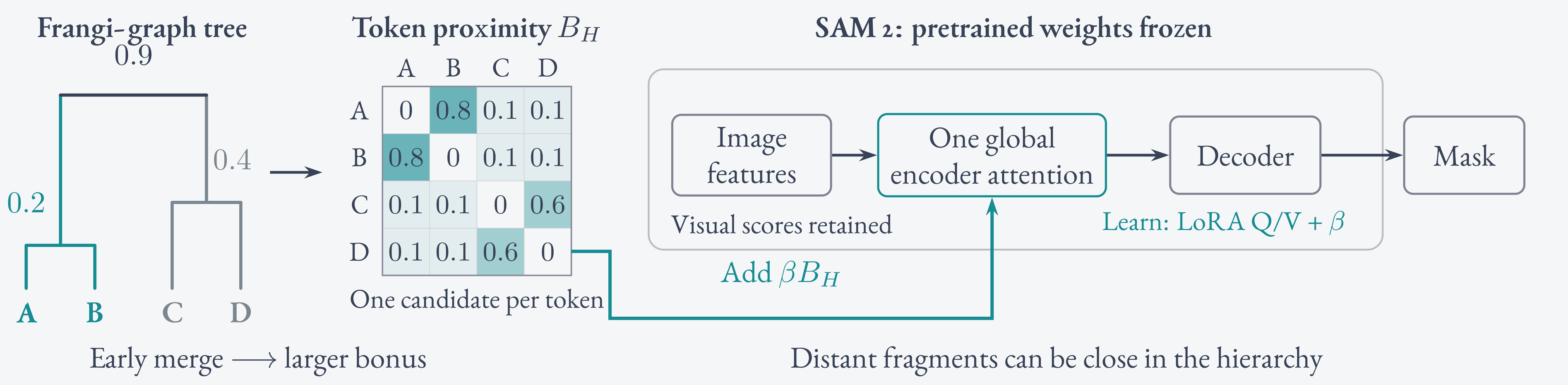

Test1: 47% / 50% Test2: 29% / 34%

Jaccard: ours / U-Net. Transfer strategy [7]; local U-Net trained on a separate image.

Ours U-Net + transfer learning Ground truth

U-Net: **about two days of expert annotation**; ours: none.

## Perspective: frozen SAM 2 + LoRA + Frangi hierarchy




| | A | B | C | D |
|---|---|---|---|---|
| A | 0 | 0.8 | 0.1 | 0.1 |
| B | 0.8 | 0 | 0.1 | 0.1 |
| C | 0.1 | 0.1 | 0 | 0.6 |
| D | 0.1 | 0.1 | 0.6 | 0 |

**SAM 2 [6]: frozen pretrained weights.**
**LoRA [9] + one coefficient $\beta$: learned together.**

$$A_H = \operatorname{softmax}(L + \beta B_H).$$

$L$: visual scores; $B_H$: hierarchy bias.
Merge heights describe connectivity [10]. We use them to bias attention, following Graphormer [8]. CrackSAM [4] motivates adaptation to cracks.
**Aim: fewer breaks without more false links.**

## References

[1] Frangi et al., *Multiscale vessel enhancement filtering*, MICCAI, 1998.
[2] Zhou et al., *FIND*, Zenodo, 2022.
[3] Jiang et al., *CrackSegDiff*, arXiv:2410.08100, 2024.
[4] Ge et al., *Fine-tuning vision foundation model for crack segmentation in civil infrastructures*, Constr. Build. Mater. 431, 136573, 2024.
[5] Zhang et al., *Deep Learning for Crack Detection: A Review of Learning Paradigms, Generalizability, and Datasets*, arXiv:2508.10256, 2025.
[6] Ravi et al., *SAM 2: Segment Anything in Images and Videos*, ICLR, 2025.
[7] Shi et al., *Semi-universal geo-crack detection by machine learning*, Front. Earth Sci. 11, 1073211, 2023.
[8] Ying et al., *Do Transformers Really Perform Badly for Graph Representation?* (Graphormer), NeurIPS, 2021.
[9] Hu et al., *LoRA: Low-Rank Adaptation of Large Language Models*, ICLR, 2022.
[10] Turaga et al., *Maximin affinity learning of image segmentation* (MALIS), NIPS, 2009.

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Code & data